Automatic Heating System

Mini Project Report

Cainen Mar · John Mark Obura · Elijah Burk

Network Analysis · Dr. Liu · March 24, 2026

Objective

Build a closed-loop heating system that automatically maintains a preset temperature using a thermistor, instrumentation amplifier, Schmitt trigger, and transistor-switched heater.

1 · Thermistor Characterization

Measurements

Condition	Temperature (°F)	Resistance (Ω)
Room temperature	74.7	569
Body temperature (trial 1)	91.0	428
Body temperature (trial 2)	89.8	442
Body temperature (average)	90.4	435

Linear Model

A linear regression on the two data points yields the characteristic equation:

$$R(T) = -8.53503\,T + 1206.56688 \quad (\Omega, °F)$$

$R^2 = 1.00$, confirming an excellent linear fit over the measured range. The model is valid for the narrow operating window of 83 °F – 91 °F.

Preset Room Temperature

Target temperature set 10 °F above true room temperature to avoid requiring active cooling:

$$T\_room = 84.7\ °F$$

2 · Schmitt Trigger Calibration

Measured Thresholds

Parameter	Value
Pin 2 (inverting) — original	−0.988 V
Pin 6 (output) — original (+Vsat)	+4.330 V
Pin 3 (non-inv.) — original (Vut)	+0.537 V
Pin 2 — flipped	+1.183 V
Pin 6 — flipped (−Vsat)	−3.480 V
Pin 3 — flipped (Vlt)	−0.431 V

Hysteresis band:

$$|Vut - Vlt| = |0.537 - (-0.431)| = 0.968\ V$$

3 · Component Value Calculations

R_1 — Thermistor Reference Resistance

Set R_1 equal to the thermistor resistance at the target temperature T = 84.7 °F:

$$R_1 = -8.53503 \times 84.7 + 1206.567 = 481\,\Omega$$

Voltage Divider Outputs at Boundary Temperatures

At T = 87 °F (upper bound — heater should be OFF):

$$R@87 = -8.53503 \times 87 + 1206.567 = 464.02\,\Omega$$

$$V2@87 = 481 / (481 + 464.02) \times 5V = 2.544\,V$$

At T = 83 °F (lower bound — heater should be ON):

$$R@83 = -8.53503 \times 83 + 1206.567 = 498.16\,\Omega$$

$$V2@83 = 481 / (481 + 498.16) \times 5V = 2.456\,V$$

Rb / Ra Ratio

The ratio of feedback resistors scales the voltage divider swing to match the Schmitt trigger hysteresis band:

$$Rb/Ra = |Vut - Vlt| / |V2@87 - V2@83|$$

$$Rb/Ra = 0.968 / (2.544 - 2.456) = 0.968 / 0.088 \approx 11$$

Selected values: Ra = 100 Ω, Rb = 1 kΩ + 82 Ω = 1082 Ω (ratio ≈ 10.82, within tolerance)

4 · Circuit Operation

The system operates as a bang-bang controller with hysteresis:

- Temperature drops below 83 °F → V2 falls below Vlt → Schmitt trigger output goes HIGH → transistor turns ON → heater activates
- Temperature rises above 87 °F → V2 rises above Vut → Schmitt trigger output goes LOW → transistor turns OFF → heater deactivates
- Hysteresis prevents rapid oscillation near a single threshold, ensuring stable on/off switching

5 · Results

Event	Expected (°F)	Measured (°F)	Error (%)
Heater OFF (upper threshold)	87.0	88.7	1.95%
Heater ON (lower threshold)	83.0	86.7	3.61%

Both errors are well within the 15% acceptable tolerance, confirming correct circuit operation. The small deviations are attributed to component tolerances, thermal lag in the enclosure, and the linear approximation of the thermistor characteristic.

Conclusion

The automatic heating system successfully regulated temperature within the target range. The thermistor-based voltage divider converted temperature to a voltage signal; the instrumentation amplifier provided high-impedance, high-gain sensing; and the Schmitt trigger with hysteresis delivered clean, stable switching. The transistor-controlled heater responded correctly at both thresholds with errors under 4%, demonstrating accurate closed-loop temperature control.